

Evidence-Grounded LLM Strategy Chains for Localized Wind Turbine Fault Operations and Maintenance

Yehong Dong and Shuai Hao

Abstract—Wind turbine operations and maintenance (O&M) relies on fault-code tables, manuals, repair records, and case reports whose meanings often depend on turbine model and field context. Plain retrieval and large-language-model (LLM) assistants can miss these scope constraints and produce answers that are difficult to audit. This paper presents a localized evidence-grounded O&M reasoning framework with three algorithmic components: scope-bound fault-code retrieval, mechanism-closure graph construction, and verification-state strategy control. The framework converts heterogeneous documents into structured fault entries, wiki pages, a domain knowledge graph, and a mechanism-enhanced reasoning graph. The LLM organizes evidence-bound troubleshooting steps rather than supplying unsupported factual content. We further analyze how scope filtering reduces ambiguous candidate sets, how mechanism closure constrains answer completeness, and how evidence-bound generation limits unsupported maintenance claims. The implementation covers 11,865 raw fault records, 22,680 graph nodes, 79,474 relations, 66 turbine models, and 29 wind farms, and achieves complete mechanism-closure and hypothesis-discrimination coverage over 33 representative fault cases.

Index Terms—Wind turbine fault diagnosis, large language model, retrieval-augmented generation, knowledge graph, operations and maintenance, local deployment, evidence traceability.

I. Introduction

WIND turbine O&M is document-intensive and context-sensitive. A field engineer receiving an alarm or a fault code must determine the applicable wind farm, turbine model, brand, control system, component, possible causes, reset conditions, and maintenance procedure before acting. These facts are usually scattered across vendor manuals, fault-code tables, field service reports, work orders, and technician experience. As wind farms become larger and more heterogeneous, this document fragmentation directly increases troubleshooting time and the risk of incorrect maintenance actions.

Conventional digital O&M tools partially address this problem through keyword search, SCADA dashboards, predictive models, or static knowledge bases. However, three practical gaps remain. First, the same numeric fault code may have different meanings across brands or

turbine models; a text-only search that ignores model scope can therefore return plausible but wrong results. Second, ordinary retrieval returns snippets but does not explicitly represent relations such as “fault code—applicable model—system—component—cause—action—source document.” Third, recent LLM-based assistants can produce fluent answers, but their answers are difficult to audit unless every key conclusion is grounded in local evidence.

This paper proposes a localized knowledge construction and evidence-grounded LLM reasoning framework for wind turbine O&M. The framework does not treat documents as independent chunks. It extracts and normalizes domain entities, builds graph relations with source references, organizes fault mechanisms into reusable templates, and generates technician-readable pages. During question answering, the LLM is used for language organization and strategy-chain control; factual claims must be supplied by retrieved local evidence.

The algorithmic contribution is threefold. First, we formulate fault-code retrieval as a scope-bound candidate selection problem in which code, turbine model, brand, system, session state, and source strength jointly determine the admissible evidence set. Second, we construct a mechanism-closure graph that requires each selected fault case to contain an explanation path, an observable, a verification test, and a prevention barrier before it is used for action recommendation. Third, we add a verification-state strategy chain that converts retrieved evidence into one field action per turn and updates the branch state from short technician feedback. This differs from plain RAG and generic GraphRAG because the retrieved unit is not only a passage or an entity community, but a scoped and falsifiable maintenance object. The evaluation combines implementation-scale statistics with four representative maintenance cases to examine auditability, mechanism closure, and field usability.

II. Related Work

A. Wind Turbine Fault Diagnosis

Wind turbine fault diagnosis has been widely studied using vibration analysis, SCADA data, signal processing, and machine learning. Yang et al. investigated SCADA-based condition monitoring for wind turbines [1], while Schlechtingen et al. proposed normal-behavior models

This work studies localized knowledge construction and retrieval-augmented reasoning for wind turbine operations and maintenance.

Yehong Dong is with the School of New Energy, North China Electric Power University. Shuai Hao is with the School of Automation Science and Electrical Engineering, Beihang University (email: yehong.dong@example.com; shuai.hao@example.com).

for fault detection [2]. Deep learning methods, including neural-network bearing diagnosis [3], autoencoders [4], LSTM models [5], and graph neural networks [6], can improve anomaly detection under sufficient labeled data. Recent work has also begun to combine visual fault detection, knowledge graphs, and LLM-based interpretation. For example, LLM-YOLOMS maps visual detection results into structured textual attributes and uses a domain-tuned LLM to produce interpretable wind-turbine component diagnosis and maintenance recommendations [7], while WT-Fault-LLM explores a knowledge-graph-enhanced large model for wind-turbine faults [8]. These methods are valuable for numerical or visual monitoring, but they do not directly solve document-level maintenance reasoning: identifying the applicable manual, interpreting model-specific fault codes, explaining evidence, and guiding a technician through multi-turn field checks.

B. Digital Twins and Industrial Knowledge Representation

Digital twin technology supports asset monitoring, simulation, and optimization [9]–[11]. In wind-power scenarios, digital twins are often connected to SCADA data and performance optimization [12]. However, O&M documents also contain causal and procedural knowledge that is difficult to represent only through numerical telemetry. Knowledge graphs offer a complementary representation because they can connect faults, components, causes, actions, and sources. Recent graph-RAG surveys further argue that graph-structured retrieval is useful when relation paths, entity communities, or global corpus structure are important to generation [13]. A 2026 systematic review of LLM-KG integration for fault diagnosis also identifies KG-enhanced LLMs and LLM-augmented KGs as emerging paradigms for improving interpretability and reducing hallucinations in industrial diagnosis [14]. The system studied here focuses on this document-grounded knowledge layer and can be integrated with digital twins or SCADA platforms as a reasoning backend.

C. Retrieval-Augmented LLMs

Retrieval-augmented generation (RAG) reduces hallucination by injecting external evidence into LLM prompts [15]. A recent survey summarizes RAG as an increasingly important paradigm for knowledge updating, long-tail information access, and cost-aware generation in AI-generated content systems [16]. However, plain chunk retrieval is insufficient for wind turbine fault handling because it does not explicitly encode model scope, fault-code ambiguity, or troubleshooting state. Self-RAG introduces adaptive retrieval and self-reflection to improve factuality [17], while corrective RAG evaluates retrieved documents and triggers alternative retrieval actions when evidence quality is low [18]. GraphRAG further constructs entity graphs and community summaries to answer global questions over private corpora [19]. Recent industrial

fault-diagnosis systems have begun to combine knowledge graphs and LLMs for diagnostic reasoning and maintenance decision support [20], and related power-system work shows that KG-guided LLM reasoning can preserve causal interpretability in converter-dominated infrastructure [21]. RAG evaluation has also become a dedicated research topic; Ragas proposes reference-free metrics for retrieval focus and generation faithfulness [22]. The proposed method follows this evidence-oriented direction but specializes it for wind-farm O&M by using domain scope constraints, fault-mechanism templates, and evidence paths rather than generic passage lists. Its strategy-chain interaction is related to reasoning-and-acting agents such as ReAct [23], but every action proposal is constrained by local evidence and field-verification criteria.

D. Recent Large Language Models for Industrial Reasoning

Recent foundation models have improved instruction following, reasoning, and tool-use capabilities. GPT-4 demonstrated strong general reasoning and multimodal competence [24]; Llama 3 showed that open-weight model families can support broad downstream adaptation [25]; Qwen2.5 expanded multilingual, coding, mathematical, and structured-data capabilities with open-weight offerings [26]; and DeepSeek-R1 highlighted the effectiveness of reinforcement learning for reasoning behavior [27]. These advances make local or enterprise-hosted LLMs more feasible for industrial settings. Nevertheless, stronger models do not remove the need for evidence control. In the proposed system, the LLM is treated as a reasoning and language layer over local O&M knowledge, not as an independent maintenance authority.

Multimodal foundation models have also accelerated industrial inspection research. GPT-4V, Claude 3, Gemini, and Qwen-VL showed that general-purpose models can combine visual understanding with textual reasoning [28]–[31]. In wind turbine inspection, LLM-YOLOMS recently combined visual detection outputs with LLM-based semantic interpretation for component fault diagnosis and maintenance support [7]. The work in this paper differs from these visual-inspection pipelines because its primary input is a private O&M document corpus and its main output is an auditable, source-grounded troubleshooting strategy.

III. System Design

A. Problem Formulation

Let $\mathcal{D}$ be the local O&M document set and $\mathcal{F} = \{f_i\}$ the extracted fault entries. Given a query q and state s_t , the goal is to select admissible evidence $\mathcal{E}_t$, construct a mechanism-closed diagnostic object y_t , and output one executable field action u_t :

$$(q, s_t, \mathcal{D}) \rightarrow \mathcal{E}_t \rightarrow y_t \rightarrow u_t. \quad (1)$$

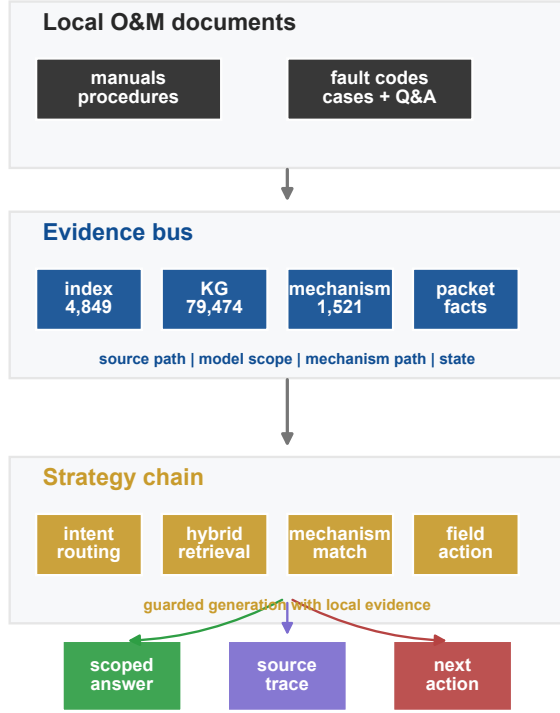


Fig. 1. Proposed system architecture. Local O&M documents are transformed into structured knowledge before they are used by the LLM. The LLM performs strategy control and answer organization, while retrieved local evidence supplies the factual basis.

The object y_t is admissible only when it contains scope, mechanism explanation, verification criterion, and source path. This makes the LLM a constrained organizer of retrieved evidence rather than an independent maintenance authority.

B. Architecture Overview

Figure 1 summarizes the architecture. Local manuals, fault-code tables, case reports, work orders, Q&A traces, and file snapshots enter through the data layer. After parsing, the knowledge layer normalizes field names, builds evidence-aware fault entries, attaches mechanism templates, and emits graph and wiki artefacts. The reasoning layer sits above this local knowledge base and handles intent routing, hybrid retrieval, graph hops, strategy-chain control, and evidence-bound answer generation. The same backend can be exposed through command-line tools, a local web service, multi-turn troubleshooting pages, evidence backtracking views, or an on-premise service.

C. Knowledge Construction Pipeline

The deterministic knowledge construction process is shown in Figure 2. The system first records document

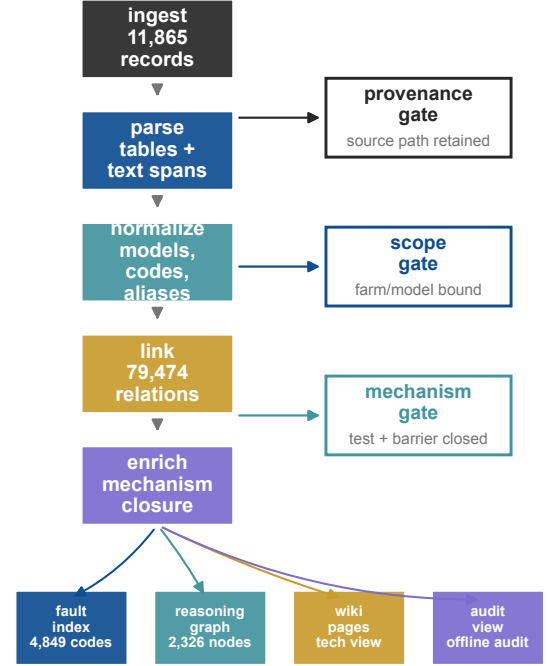


Fig. 2. Knowledge construction flow from local document ingestion to graph, mechanism, index, and wiki outputs.

provenance, extracts text and table content, identifies domain fields, normalizes aliases, and creates graph relations. It then attaches fault entries to mechanism templates and generates wiki pages and retrieval indexes.

The knowledge schema is intentionally practical rather than overly broad. It includes wind farms, turbine models, brands, fault codes, systems, components, causes, recommended actions, reset methods, mechanism templates, and source documents. For every extracted fault entry, the system keeps both normalized fields and original source metadata. This design supports retrieval and human audit simultaneously.

Formally, a fault entry is represented as

$$f_i = (c_i, m_i, b_i, s_i, p_i, r_i, a_i, e_i), \quad (2)$$

where c_i is the fault code or alarm name, m_i is the applicable turbine model, b_i is the brand or controller family, s_i is the system, p_i is the component, r_i is the cause description, a_i is the action or reset rule, and e_i is the evidence pointer. The evidence pointer stores the source file, local path, section or page hint when available, and the raw extracted text span. A mechanism template is represented as

$$g_j = (u_j, v_j, q_j, k_j), \quad (3)$$

where u_j denotes typical symptoms, v_j denotes causal factors, q_j denotes priority checks, and k_j denotes acceptance criteria or branch conditions. The mapping $f_i \rightarrow g_j$ is created from system type, component, fault manifestation, cause terms, and recommended actions. This mapping is intentionally many-to-one because different fault codes may share the same field mechanism.

D. Domain Relationship Model

A fault-code query is not interpreted only as a number. The system first binds the query to wind-farm or turbine-model context whenever available. If the same code appears in several model scopes, it asks for disambiguating information or returns separate scoped candidates. This avoids the common failure mode where a generic answer is correct for one turbine type but wrong for the turbine being repaired.

The scope-binding score for a candidate fault entry combines exact fault-code match, model compatibility, brand compatibility, system-term overlap, and session-context consistency:

$$\text{score}(f_i, q) = \lambda_c C_i + \lambda_m M_i + \lambda_b B_i + \lambda_s S_i + \lambda_h H_i + \lambda_e E_i, \quad (4)$$

where C_i is the fault-code match signal, M_i and B_i measure model and brand consistency, S_i measures system/component overlap, H_i measures consistency with previous conversation state, and E_i rewards entries with stronger source evidence. If several high-scoring candidates have incompatible model scopes, the system reports the ambiguity and requests a missing scope condition instead of collapsing the results into a single answer.

This module is a constrained ranking operator. Let $\mathcal{F}(c)$ be entries sharing a code c , and let $\Omega(q, s_t)$ be the inferred scope constraints. The admissible candidate set is

$$\mathcal{A}(q, s_t) = \{f_i \in \mathcal{F}(c) : f_i \models \Omega(q, s_t)\}. \quad (5)$$

When $\Omega(q, s_t)$ is non-empty, $\mathcal{A}(q, s_t) \subseteq \mathcal{F}(c)$; thus scope binding cannot enlarge fault-code ambiguity. If the admissible set is empty or scope-incompatible, the algorithm withholds action and requests missing scope information.

E. Fault-Mechanism Templates

Individual fault records are often too fragmented for field diagnosis. The system therefore maps records to mechanism templates, such as power-supply feedback-chain faults, pitch actuator blocking, converter DC-link or grid-voltage disturbance, yaw brake and hydraulic-loop abnormalities, safety-chain interruption, analog-signal interference, and bearing lubrication-wear coupling. Each template records typical symptoms, evidence, priority checks, acceptance criteria, and recommended actions. The answer can then explain not only “what the code means” but also “which mechanism is likely” and “what should be verified next.”

F. Mechanism-Enhanced Reasoning Graph

To make the knowledge graph reason at the mechanism level rather than only at the symptom-action level, this work constructs an additional reasoning graph over fault cases. The core node types are mechanism archetype, mechanism layer, failure mode, propagation step, observable, verification test, control barrier, diagnostic hypothesis, discriminating evidence, counterfactual test, and decision rule. Six reusable mechanism archetypes are currently used: hydraulic energy and flow restriction, load-lubrication contact fatigue, electrical thermal and insulation stress, sensing-acquisition-control feedback, communication-link timing and data consistency, and protection-chain boundary judgment.

For each fault case, the graph first matches one or two mechanism archetypes using anchor terms, component context, symptoms, cause descriptions, and neighboring graph entities. It then instantiates a closed reasoning chain:

$$x \rightarrow a \rightarrow l \rightarrow p \rightarrow z \rightarrow o \rightarrow t \rightarrow b, \quad (6)$$

where x is a fault case, a is a mechanism archetype, l is a mechanism layer, p is a propagation step, z is a failure mode, o is an observable, t is a verification test, and b is a control barrier. This chain allows the system to answer why a signal should be checked, how the bottom-level mechanism propagates to the observed alarm, and how recurrence can be prevented.

A second layer is added for diagnostic discrimination. If a case matches two mechanism archetypes, the system creates a pairwise competing hypothesis, such as distinguishing real hydraulic flow restriction from pressure-sensor or acquisition-chain errors. If a case has only one dominant archetype, the system creates a single-mechanism counterfactual hypothesis that distinguishes the true physical mechanism from measurement artifacts, boundary-condition effects, or control-timing pseudo-causes. Each hypothesis is connected to discriminating evidence, counterfactual tests, and decision rules:

$$x \rightarrow h \rightarrow \{e_d, t_c, r_d\}, \quad (7)$$

where h is the diagnostic hypothesis, e_d is discriminating evidence, t_c is a counterfactual test, and r_d is the decision rule. This design makes the graph support falsifiable troubleshooting rather than only ranked retrieval.

G. LLM Strategy Chain for Troubleshooting

Unlike a one-shot answer, the strategy chain keeps a state consisting of the current fault object, model scope, retrieved evidence, completed verification actions, user feedback, and remaining branches. It then recommends one primary field action per turn.

The strategy chain enforces an answer contract with six fields: fault meaning, applicable scope, likely mechanism, one verification action, acceptance criterion, and evidence path. This contract is designed for field execution. For example, when a yaw hydraulic pressure fault is retrieved,

the answer should not merely list all possible hydraulic components. It should first verify whether pressure drops sharply during brake release, specify the expected recovery time, and then branch based on whether the measured recovery is normal. The state object stores the selected fault entry, the last action, the expected observation, and the user's feedback so that a short reply can be interpreted as a branch result rather than a new unrelated question.

H. Theoretical Properties

The framework provides three bounded properties for field O&M. First, scope filtering reduces the search space before generation: $|\mathcal{A}(q, s_t)| \leq |\mathcal{F}(c)|$. Second, mechanism closure defines a completeness constraint:

$$\phi(y) = I_{\text{scope}}(y)I_{\text{mech}}(y)I_{\text{test}}(y)I_{\text{barrier}}(y)I_{\text{source}}(y), \quad (8)$$

where each indicator is one only when the corresponding field is present. The system recommends an action only when $\phi(y) = 1$, forcing explanation, verification, prevention, and source traceability to appear together. Third, evidence-bound generation requires each maintenance-relevant factual claim in the answer to be supported by an element of $\mathcal{E}_t$, except for explicit clarification or safety wording. These properties constrain the prompt and output space; they do not prove that the selected mechanism is always correct.

IV. Implementation

A. Local Knowledge Assets

The current prototype uses a file-based knowledge store for reproducibility and simple offline deployment. The generated artefacts include a fault index for model-scoped lookup, a knowledge graph for entity-relation browsing, a reasoning graph for causal and procedural relations, Mark-down wiki pages for technician-facing reading, and HTML graph views for offline inspection and demonstration. This file-oriented design is not intended to replace database-backed deployment; rather, it keeps the construction process transparent during development and makes the evidence paths easy to inspect.

Figure 3 reports the current scale of the knowledge base used in this study. The system contains 11,865 raw fault records, 22,680 graph nodes, 79,474 relations, 4,849 fault-code nodes, and 8,295 source-document nodes. It covers 13 brands, 66 turbine models, 29 wind farms, 18 systems, and 12 mechanism templates.

B. Retrieval and Answer Generation

Question answering is implemented as a guarded generation process rather than an unconstrained chat interface. The system first decides whether the user question belongs to a wind-power fault scenario by considering fault codes, domain terms, action verbs, and the conversation context. Non-fault questions can be handled by a general assistant, while fault questions enter the local knowledge route. In that route, the query is bound to wind

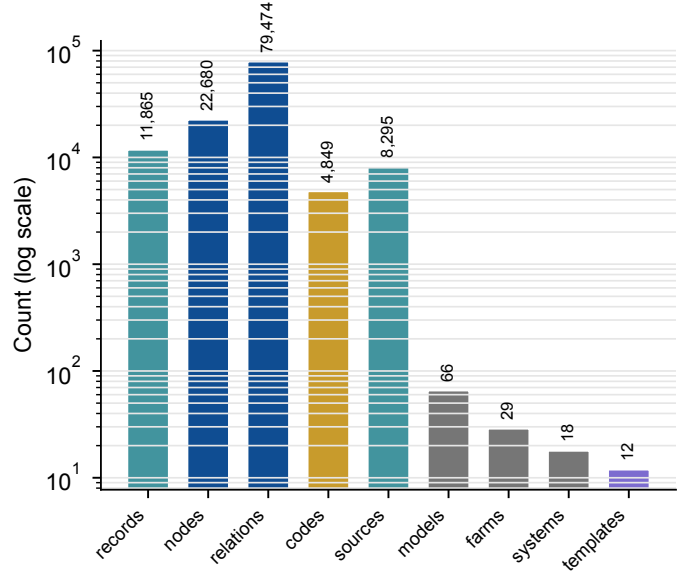


Fig. 3. Current knowledge-base scale. Counts are shown on a logarithmic axis to make document, graph, and mechanism resources comparable.

farm, turbine model, brand, system, or component scope whenever such information is available. The retriever then collects structured fault entries, mechanism templates, graph neighbors, and source snippets, and these elements are compressed into an evidence packet containing only auditable facts and source identifiers.

The LLM receives this evidence packet together with the retained troubleshooting state. Its answer is expected to cover the fault meaning, applicable scope, likely causes, priority checks, acceptance criteria, recommended actions, and evidence path, but the factual content is constrained by retrieved local knowledge. When the evidence is insufficient or mutually incompatible, the system returns a clarification request instead of a definitive maintenance instruction. This design is deliberately conservative because field O&M answers often lead to physical operations on equipment.

C. Local and Intranet Deployment

The system is designed for local workstations, edge computers, or enterprise intranet servers. Sensitive O&M documents do not need to leave the operator boundary. The text-generation model may be a local model service, an enterprise-hosted model endpoint, or an approved cloud API, but the knowledge construction, evidence retrieval, and source backtracking remain local. This deployment model is important for wind farms with weak network connectivity or strict data-governance policies.

V. Case Studies and Evaluation

A. Evaluation Perspective

The evaluation focuses on three questions that matter in field deployment. The first is coverage: whether the knowledge base preserves enough model, fault, source, and

mechanism entities for practical retrieval. The second is auditability: whether a generated answer can be traced back to local evidence rather than unsupported LLM memory. The third is field usability: whether the answer is organized as executable verification actions instead of a long unstructured explanation.

The local case studies are therefore read together with a conceptual comparison against recent RAG and graph-RAG methods. Plain RAG improves knowledge access but remains passage-centric [15], [16]. Self-RAG and CRAG add retrieval control and critique loops [17], [18]. GraphRAG incorporates structured entity relations and community summaries [19]. The proposed system follows the same movement toward evidence-aware generation, but it is tuned for industrial fault handling, where model scope, source path, and stepwise field verification are more important than free-form summarization quality.

B. Knowledge-Base Scale

Table I summarizes the current implementation scale. These counts are generated from local project assets and are used as a sanity check for retrieval coverage.

TABLE I
Knowledge-Base Scale

Item	Count
Raw fault records	11,865
Graph nodes	22,680
Graph relations	79,474
Fault-code nodes	4,849
Source-document nodes	8,295
Brands	13
Turbine models	66
Wind farms	29
Systems	18
Mechanism templates	12

C. Mechanism-Graph Evaluation

Table II reports the structural evaluation of the mechanism-enhanced reasoning graph. A mechanism-closed case is defined as one that has a mechanism archetype, failure mode, observable, verification test, and control barrier. A hypothesis-discriminated case is defined as one that has a diagnostic hypothesis, discriminating evidence, counterfactual test, and decision rule. These metrics measure whether the graph can support both explanation and falsification.

The results show that all 33 representative cases are connected to complete mechanism-verification-prevention chains. Compared with the traditional fault-evidence-action profile, whose completeness is 72.7%, the mechanism-enhanced profile reaches 100.0% because each case has explicit mechanism terms, failure-mode terms, and verification terms. The hypothesis layer also reaches 100.0% coverage: 18 cases are handled by pairwise mechanism competition and 15 cases by single-mechanism counterfactual discrimination. This improves the graph's

TABLE II
Mechanism-Enhanced Reasoning Graph Statistics

Item	Value
Reasoning-graph nodes	2,326
Reasoning-graph relations	3,567
Mechanism nodes	1,521
Mechanism relations	2,436
Fault cases	33
Mechanism archetypes	6
Mechanism-closure coverage	100.0%
Hypothesis-discrimination coverage	100.0%
Pairwise mechanism-discrimination cases	18
Single-mechanism counterfactual cases	15
Traditional profile completeness	72.7%
Mechanism-profile completeness	100.0%

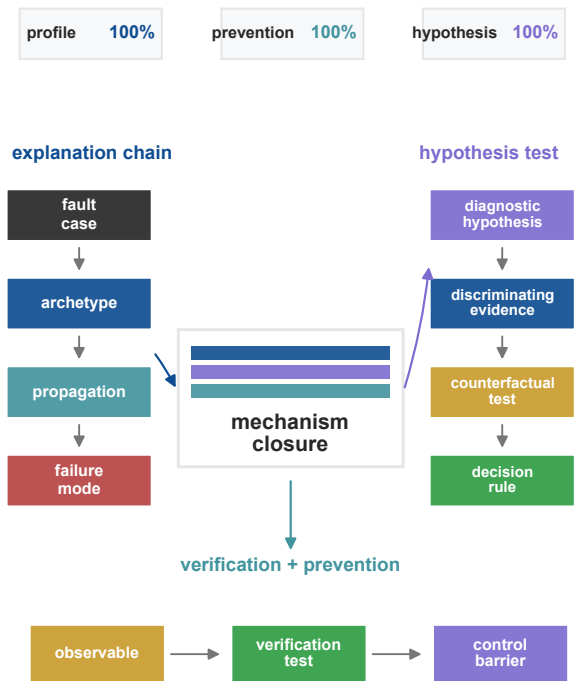


Fig. 4. Mechanism-enhanced reasoning graph as a single closed diagnostic chain. The figure combines the fault-to-mechanism path, the counterfactual hypothesis branch, and the closure gains over a traditional fault-evidence-action profile.

ability to recommend not only a likely cause but also the evidence required to reject plausible alternatives.

The fault-code index also exposes why scope binding is needed. Among 4,849 unique fault-code nodes, 1,714 codes map to more than one brand/model/fault-name group, and the most ambiguous code maps to 19 scoped meanings. A bare code lookup or plain chunk retriever can therefore mix incompatible maintenance procedures unless the model and brand scope are represented explicitly.

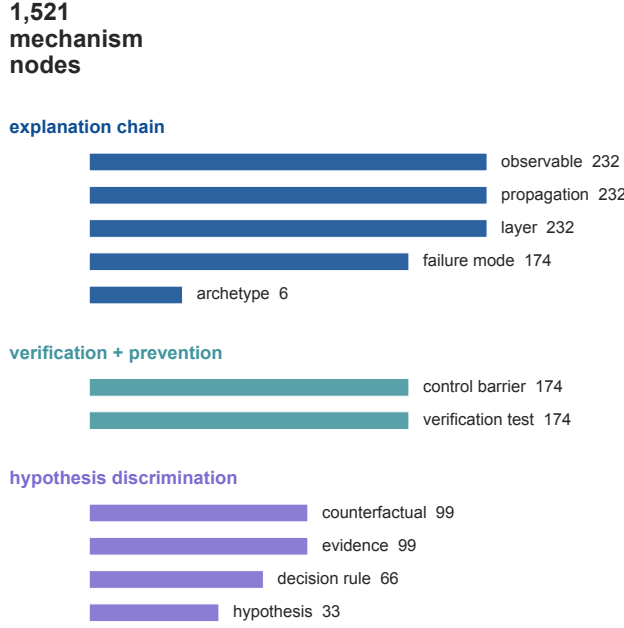


Fig. 5. Composition of the mechanism-enhanced reasoning graph. Node groups are arranged radially around the mechanism core to show how mechanism layers, propagation steps, observables, tests, barriers, hypotheses, evidence, counterfactual tests, and decision rules contribute to the graph.

D. Ablation and Case-Level Analysis

Figure 4 gives a structural ablation between the traditional fault-evidence-action profile and the mechanism-enhanced reasoning graph. The traditional profile already supports validation actions because it contains diagnostic signals and maintenance steps, but it does not explicitly encode recurrence-prevention barriers or falsifiable diagnostic hypotheses. After adding the mechanism layer, profile completeness increases from 72.7% to 100.0%, prevention closure increases from 0 to 100.0%, and hypothesis-discrimination closure increases from 0 to 100.0%.

Figure 5 further decomposes the mechanism graph by node and relation type. The largest node groups are mechanism layers, propagation steps, and observables, each with 232 instances, followed by failure modes, verification tests, and control barriers, each with 174 instances. The diagnostic-hypothesis layer contributes 33 hypotheses, 99 discriminating-evidence nodes, 99 counterfactual-test nodes, and 66 decision-rule nodes. This distribution indicates that the added graph is not only a small set of labels attached to fault cases; it expands each case into a mechanism, observation, verification, prevention, and discrimination structure.

Figure 6 reports case-level quality. The average mechanism path depth is 3.00 and the maximum depth is 3, meaning that each case has a consistent multi-hop path from fault case to mechanism evidence rather than an iso-

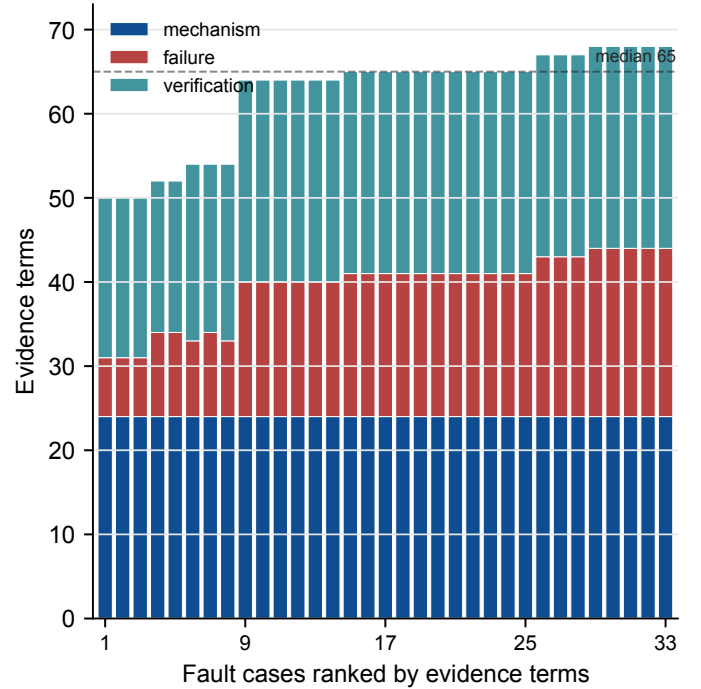


Fig. 6. Per-case mechanism quality after graph enrichment. Each bar is one fault case, coloured by hypothesis type, showing that all evaluated cases obtain a complete mechanism-verification-prevention chain.

lated tag. All 33 cases have observables, verification tests, and control barriers, which is important for generating field actions that can be checked and audited.

Figure 7 shows that the gain is not limited to more graph edges: the added layer specifically fills prevention and hypothesis-discrimination dimensions that are missing in the traditional profile. The graph also spans six reusable physical reasoning modes, and Table II shows that 18 cases require pairwise competition between two mechanism archetypes whereas 15 cases use a single-mechanism counterfactual branch. Thus, the proposed graph does not only retrieve a likely cause; it records how that cause can be separated from plausible alternatives.

E. Representative Troubleshooting Cases

Table III lists four representative cases extracted from local wind-power fault documents. Each case contains a causal chain and recommended action sequence that can be retrieved and converted into strategy-chain guidance.

The case library is intentionally heterogeneous. The first two cases are especially useful because they demonstrate both mechanism-level reasoning and source-supported field action. The IGBT-board case emphasizes environmental and thermal contributors. The yaw-pressure case emphasizes hidden hydraulic flow restriction and recovery timing, which is a stronger test of multi-step reasoning than a simple binary alarm lookup. The bearing case introduces coupled wear factors that require condition-monitoring evidence. The temperature-jump case demonstrates why source evidence matters: a scalar temperature

TABLE III
Representative Evidence-Grounded Fault Cases

Case	Mechanism chain	Key evidence	Recommended action
Converter board fault	IGBT-Grid-voltage fluctuation, dust, salt fog, humidity, or poor heat dissipation leads to IGBT driver-board short circuit, capacitor damage, high temperature, and converter trip.	ISU SC and INU SC alarms; K8 contactor cannot close; temperature sensor not abnormal; post-retrofit operating temperature decreases.	Replace IGBT driver board; clean filters; improve ventilation and dehumidification; monitor board temperature.
Yaw hydraulic pressure abnormality	Hidden damping buffer blockage restricts yaw hydraulic-loop oil flow, causing slow pressure recovery and hydraulic-station undervoltage.	Manual yaw-brake release drops pressure from 151 bar to 27 bar; pressure recovery to 150 bar takes about 300 s; after cleaning, pressure rises to 153 bar in 11 s.	Clean plunger holes and damping buffer; adjust buffer opening; replace with improved buffer; update hydraulic drawings.
Main-shaft bearing composite wear	High-altitude temperature variation, turbulent load, lubrication contamination, coaxiality deviation, and seal aging cause oil-film rupture and spalling.	Vibration velocity 4.8 mm/s; kurtosis 8.7; metal debris 1200 ppm; coaxiality deviation 0.15 mm; inner and outer race spalling.	Stop turbine; replace bearings; clean lubrication chamber; repair shaft journal; upgrade seal and online oil filtration.
Generator winding temperature jump	Poor shielding, grounding faults, irregular cable routing, and electromagnetic coupling distort Pt100 analog temperature signals.	Temperature jumps between 70–135 °C; stable when stopped; fluctuation increases after grid connection; channel swap excludes PLC module fault.	Check Pt100 resistance; swap acquisition channel; fix shielding and grounding; replace signal cable; switch backup sensor if needed.

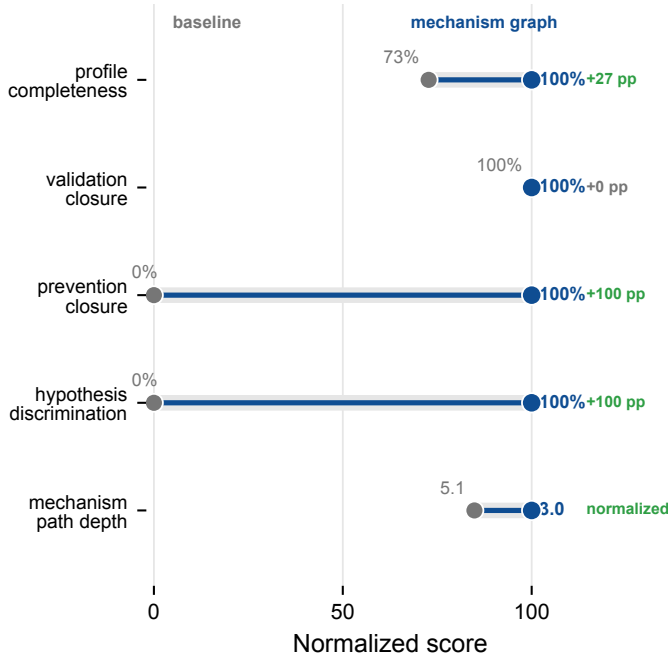


Fig. 7. Ablation matrix comparing the traditional fault-evidence-action profile with the mechanism-enhanced reasoning graph. The mechanism layer adds prevention closure, hypothesis discrimination, and a normalized mechanism path-depth measure while preserving validation closure.

value alone is not sufficient without channel, grounding, and load context.

F. Module Ablation

Table IV reports a module-level ablation over the fault-code index and the 33 representative mechanism cases. The metrics are structural readiness measures: ambiguous-code control measures whether ambiguous codes are rep-

resented as scoped candidates instead of collapsed into a single answer; profile, prevention, and hypothesis coverage measure whether the corresponding graph fields exist; and strategy readiness requires discriminating evidence, a verification test, and a decision rule.

TABLE IV
Module Ablation of Algorithmic Components

Variant	Amb.	Prof.	Prev.	Hyp.	Strat.
Plain chunk RAG	0.0	72.7	0.0	0.0	0.0
Scope only	100.0	72.7	0.0	0.0	0.0
Scope + closure	100.0	100.0	100.0	0.0	0.0
Full framework	100.0	100.0	100.0	100.0	100.0

Amb.: scoped control of ambiguous fault codes; Prof.: profile completeness; Prev.: prevention closure; Hyp.: hypothesis-discrimination coverage; Strat.: verification-strategy readiness. Values are percentages.

The ablation separates the roles of the three proposed modules. Scope-bound retrieval controls fault-code ambiguity but does not by itself add prevention barriers or falsifiable hypotheses. Mechanism closure raises profile completeness and prevention closure to 100.0%, but strategy readiness remains zero until the hypothesis and decision-rule layer is added. The full framework is therefore needed to produce a scoped, mechanism-closed, and verification-ready diagnostic object.

VI. Discussion

A. Why Structured Local Knowledge Matters

The main engineering lesson is that industrial O&M assistants should not rely on LLM memory or plain text retrieval alone. Wind turbine fault handling depends on scope constraints, mechanism discrimination, and evidence paths. The proposed algorithmic design addresses these requirements before generation: the scope-bound

retriever reduces incompatible fault-code candidates, the mechanism-closure graph enforces explanation and verification fields, and the strategy chain converts the selected diagnostic object into one executable field action. The LLM is still useful, but its role is shifted from “inventing a diagnosis” to “organizing retrieved facts into an executable troubleshooting strategy.”

The mechanism-enhanced graph further changes the reasoning target from fault matching to mechanism testing. In field diagnosis, several explanations may produce similar alarms. A pressure alarm, for example, may come from real hydraulic flow restriction, a blocked damping element, a pressure sensor error, or a control-sequence artifact. The proposed hypothesis layer represents these alternatives explicitly and attaches the evidence that can separate them. This is important because a maintenance recommendation is more useful when it states what observation would falsify it.

This design is aligned with recent RAG directions. Retrieval quality, not only model size, determines answer trustworthiness in private corpora [16], [22]. Graph-based retrieval helps when multiple documents must be connected through shared entities or causal paths [13], [19]. Self-reflection and corrective retrieval are useful in open-domain systems [17], [18], but the wind-farm setting benefits more from deterministic scope filters and auditable source paths than from unconstrained iterative exploration.

The theoretical analysis also clarifies the boundary of the contribution. The proposed constraints reduce the admissible search space and force completeness of the generated diagnostic object; they do not prove that the selected mechanism is always correct. In practice, the benefit is strongest when local documents contain reliable model scope, source paths, and field-verification criteria. When documents are incomplete or contradictory, the conservative behavior should be clarification or escalation rather than confident generation.

B. Auditability and Human Review

Evidence traceability is essential in field maintenance because a wrong instruction can damage equipment or prolong downtime. The proposed system keeps source-document nodes and evidence packets so that a technician or specialist can verify key conclusions. This also supports incremental correction: if an extracted record is wrong, the corresponding wiki page, graph edge, or source mapping can be fixed without redesigning the entire assistant.

C. Limitations

The current system is strongest for document-grounded reasoning and procedural fault handling. It does not replace sensor-based anomaly detection, vibration analysis, or expert sign-off for high-risk operations. Some source documents are scanned PDFs or have complex two-column layouts, which can introduce extraction noise. Future work will integrate OCR quality checks, larger

field evaluations, SCADA signal links, and automatic contradiction detection across manuals and repair records.

Another limitation is evaluation granularity. The present manuscript reports implementation scale and representative case studies, but a larger benchmark with held-out wind-farm scenarios would be needed to quantify retrieval precision, answer faithfulness, and repair-time reduction under controlled experimental conditions. That benchmark should include fault-code ambiguity across models, noisy technician feedback, and partial evidence availability.

VII. Conclusion

This paper presented a localized evidence-grounded LLM framework for wind turbine O&M. Its core algorithmic contribution is the combination of scope-bound fault-code retrieval, mechanism-closure graph construction, and verification-state strategy control. The framework converts local maintenance documents into structured fault entries, knowledge graphs, wiki pages, and mechanism templates, then uses an LLM strategy chain to generate auditable troubleshooting guidance. The theoretical analysis shows how the framework reduces admissible fault-code candidates, constrains action generation by mechanism completeness, and ties maintenance-relevant claims to retrieved evidence. The implementation demonstrates that deterministic knowledge construction and constrained LLM generation can support fault-code disambiguation, evidence backtracking, and multi-turn field interaction in wind-farm O&M.

Acknowledgment

This work was partially supported by National Natural Science Foundation of China under grant U25B20226.

References

- [1] W. Yang, R. Court, and J. Jiang, “Wind turbine condition monitoring by the approach of scada data analysis,” *Renewable Energy*, vol. 53, pp. 365–376, 2009.
- [2] M. Schlechtingen, I. F. Santos, and S. Achiche, “Wind turbine condition monitoring based on scada data using normal behavior models. part 1: System description,” *Applied Soft Computing*, vol. 13, no. 1, pp. 259–270, 2013.
- [3] P. Bangalore and L. B. Tjernberg, “An artificial neural network approach for early fault detection of gearbox bearings,” *IEEE Transactions on Smart Grid*, vol. 6, no. 2, pp. 980–987, 2015.
- [4] H. Shao, H. Jiang, H. Zhao, and F. Wang, “A novel deep autoencoder feature learning method for rotating machinery fault diagnosis,” *Mechanical Systems and Signal Processing*, vol. 95, pp. 187–204, 2018.
- [5] J. Chen, Y. Wang, and L. Zhang, “Fault diagnosis of wind turbine based on long short-term memory networks,” *Renewable Energy*, vol. 146, pp. 1–8, 2020.
- [6] Y. Li, X. Wang, and Z. Zhang, “Graph neural networks for wind turbine fault diagnosis,” *Applied Energy*, vol. 308, p. 118373, 2022.
- [7] Y. Li, Y. Wang, M. Li, X. Li, and J. Feng, “LLM-YOLOMS: Large language model-based semantic interpretation and fault diagnosis for wind turbine components,” *arXiv preprint arXiv:2511.10394*, 2025.
- [8] F. Wang, X. Zhang, R. Tang, and J. Jiang, “WT-Fault-LLM: Knowledge graph enhanced large model for wind-turbine faults,” in *Procedia Computer Science*, vol. 266. Elsevier, 2025, pp. 385–390.

- [9] M. Grieves and J. Vickers, *Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems*. Springer, 2017.
- [10] F. Tao, H. Zhang, A. Liu, and A. Y. Nee, "Digital twin in industry: State-of-the-art," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 4, pp. 2405–2415, 2019.
- [11] B. R. Barricelli, E. Casiraghi, and D. Fogli, "A survey on digital twin: Definitions, characteristics, applications, and design implications," *IEEE Access*, vol. 7, pp. 167 653–167 671, 2019.
- [12] K. Liu, Z. Chen, J. Wu, and S. Cheng, "A digital twin-based approach for optimization of wind farm energy production," *Energy Conversion and Management*, vol. 238, p. 114144, 2021.
- [13] B. Peng, Y. Zhu, Y. Liu, X. Bo, H. Shi, C. Hong, Y. Zhang, and S. Tang, "Graph retrieval-augmented generation: A survey," *arXiv preprint arXiv:2408.08921*, 2024.
- [14] W. B. M. Razaq, H. Chen, M. R. Machado, and J. L. Rebelo Moreira, "How can the integration of AI large language models and knowledge graph enhance fault diagnosis? a systematic literature review," *Applied Soft Computing*, vol. 194, p. 114908, 2026.
- [15] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.
- [16] P. Zhao, H. Zhang, C. Yu, X. Wang, Y. Geng, F. Fu, L. Yang, W. Zhang, and B. Cui, "Retrieval-augmented generation for ai-generated content: A survey," *arXiv preprint arXiv:2402.19473*, 2024.
- [17] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-rag: Learning to retrieve, generate, and critique through self-reflection," *arXiv preprint arXiv:2310.11511*, 2023.
- [18] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective retrieval augmented generation," *arXiv preprint arXiv:2401.15884*, 2024.
- [19] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, "From local to global: A graph rag approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024.
- [20] Y. Ma, S. Zheng, Z. Yang, H. Pan, and J. Hong, "A knowledge-graph enhanced large language model-based fault diagnostic reasoning and maintenance decision support pipeline towards industry 5.0," *International Journal of Production Research*, pp. 1–22, 2026.
- [21] Y. Lan, M. Zhang, M. Su, and F. Zhou, "Knowledge-graph-enhanced and LLM-guided fault diagnosis for VSC-HVDC systems," *AIP Advances*, vol. 15, no. 11, p. 115330, 2025.
- [22] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "Ragas: Automated evaluation of retrieval augmented generation," *arXiv preprint arXiv:2309.15217*, 2023.
- [23] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafraan, K. Narasimhan, and Y. Cao, "React: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations*, 2023.
- [24] OpenAI, "Gpt-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023.
- [25] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan et al., "The llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.
- [26] Qwen Team, "Qwen2.5 technical report," *arXiv preprint arXiv:2412.15115*, 2024.
- [27] DeepSeek-AI, "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning," *arXiv preprint arXiv:2501.12948*, 2025.
- [28] OpenAI, "Gpt-4v(ision) system card," OpenAI, Tech. Rep., 2023.
- [29] Anthropic, "Introducing claude 3," Anthropic Blog, 2024.
- [30] G. DeepMind, "Gemini: A family of highly capable multimodal models," Google, Tech. Rep., 2023.
- [31] J. Bai, S. Bai, S. Yang, S. Wang, S. Tan, P. Wang, J. Lin, C. Zhou, and J. Zhou, "Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond," *arXiv preprint arXiv:2308.12966*, 2023.



Yehong Dong is a professor with the School of New Energy, North China Electric Power University. He received the Ph.D. degree from Tsinghua University, Beijing, China, in 2011. His current research interests include wind turbine design and maintenance, machine learning, and AI-enhanced methods for failure diagnosis of wind power devices.



Shuai Hao received the B.S. degree from Beijing Institute of Technology, Beijing, China, in 2018, and the M.S. degree from Beihang University, Beijing, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Automation Science and Electrical Engineering, Beihang University. His current research interests include deep reinforcement learning and unmanned aerial vehicle cooperative control.